Gebze Technical University

Department of Mechanical Engineering

Recitation 4

Damped Free Vibration

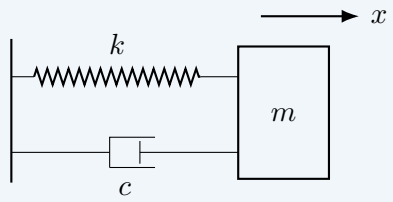
Lecturer: Saeed Lotfan | saeedlotfan.com

Self-study guide. These 30 questions develop the Week 4 topics through explanations and worked calculations. Each question is followed immediately by its solution.

Levels: Foundation builds the basic relations; Application uses them in a specified model; Challenge combines several steps or requires closer interpretation. The questions progress within each topic. Angles are in radians unless degrees are stated. Use $g = 9.81$ m/s^2.

Question 1 | From damping force to response parameters — Foundation

A mass m is connected to a linear spring k and a viscous damper c. Displacement x is measured from static equilibrium. Derive the free-motion equation, define ω_n, c_c, and ζ, and explain how the response type follows from the characteristic equation. Assume $m, k > 0$ and $c \geq 0$.



Solution 1

1. Balance incremental forces

The spring force is $-kx$ and the damping force is $-c\dot{x}$. Constant loads have already been balanced at static equilibrium. Thus

$$m\ddot{x} = -kx - c\dot{x} \quad \Rightarrow \quad m\ddot{x} + c\dot{x} + kx = 0.$$

2. Normalize and define the parameters

Divide by m and compare with $\ddot{x} + 2\zeta\omega_n\dot{x} + \omega_n^2 x = 0$:

$$\boxed{\omega_n = \sqrt{k/m}, \quad c_c = 2m\omega_n = 2\sqrt{km}, \quad \zeta = c/c_c.}$$

ω_n has units rad/s, c_c has units N s/m, and ζ is dimensionless.

3. Use an exponential trial response

Putting $x = Ce^{st}$ gives $s^2 + 2\zeta\omega_n s + \omega_n^2 = 0$, so

$$s_{1,2} = -\zeta\omega_n \pm \omega_n\sqrt{\zeta^2 - 1}.$$

For $0 < \zeta < 1$, the roots are complex and motion oscillates while decaying. At $\zeta = 1$ they coincide: critical damping. For $\zeta > 1$ they are distinct negative real numbers: overdamping. When $c = 0$, the motion is undamped.

Question 2 | Three parameter sets — Foundation

For each system below, find ω_n, c_c, ζ, and the response type. Give an oscillation frequency only when oscillation exists.

	m (kg)	c (N s/m)	k (N/m)
(a)	5	500	5000
(b)	5	500	50000
(c)	5	1000	50000

Solution 2

1. Apply the definitions

Use $\omega_n = \sqrt{k/m}$, $c_c = 2\sqrt{km}$, and $\zeta = c/c_c$:

	ω_n (rad/s)	c_c (N s/m)	ζ
(a)	31.6228	316.228	1.58114
(b)	100	1000	0.5
(c)	100	1000	1

2. Interpret each result

(a) is overdamped. Its equation is $\ddot{x} + 100\dot{x} + 1000x = 0$, with roots -11.2702 and -88.7298 s^{-1}. There is no periodic oscillation; do not report an imaginary number as a physical oscillation frequency.

(b) is underdamped:

$$\boxed{\omega_d = 100\sqrt{1 - 0.5^2} = 86.6025 \text{ rad/s}.}$$

(c) is critically damped. Its repeated root is $s = -100$ s^{-1}, and $x = (C + Dt)e^{-100t}$. Although the expression $\omega_n\sqrt{1-\zeta^2}$ becomes zero, this does not describe a zero-frequency periodic vibration.

Question 3 | Initial conditions and damped phase — Application

For $m = 1$ kg, $c = 5$ N s/m, and $k = 16$ N/m, determine the response for (a) $x_0 = 0.1$ m, $v_0 = 2$ m/s and (b) $x_0 = -0.1$ m, $v_0 = 2$ m/s. Express the phase using $x = Ae^{-\zeta\omega_n t}\cos(\omega_d t - \phi)$.

Solution 3

1. Calculate common parameters

$$\omega_n = 4, \quad c_c = 8, \quad \zeta = 0.625, \quad a = \zeta\omega_n = 2.5\ \text{s}^{-1}, \quad b = \omega_d = \sqrt{9.75} = 3.122499\ \text{rad/s}.$$

2. Impose both initial conditions

Write $x = e^{-at}(C\cos bt + D\sin bt)$. At $t = 0$,

$$C = x_0, \qquad v_0 = -aC + bD \quad \Rightarrow \quad D = \frac{v_0 + ax_0}{b}.$$

Therefore, with t in seconds and x in meters,

$$\boxed{x_a = e^{-2.5t}[0.1\cos(3.122499t) + 0.72057669\sin(3.122499t)]},$$

$$\boxed{x_b = e^{-2.5t}[-0.1\cos(3.122499t) + 0.56044854\sin(3.122499t)]}.$$

3. Convert without losing the quadrant

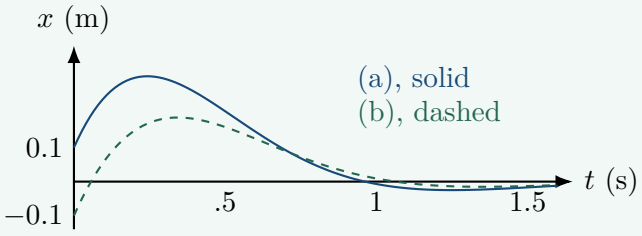
Since $C = A\cos\phi$ and $D = A\sin\phi$,

$$A = \sqrt{C^2 + D^2}, \qquad \phi = \operatorname{atan2}(D, C).$$

For (a), $A = 0.72748249$ m and $\phi = 1.43289936$ rad.
For (b), $A = 0.56930007$ m and $\phi = 1.74736668$ rad, in quadrant II.

4. Check the derivative

Case (a): $-2.5(0.1) + 3.122499(0.72057669) = 2$.
Case (b): $-2.5(-0.1) + 3.122499(0.56044854) = 2$.
The exponential decay rate is 2.5, and the damped phase requires $v_0 + ax_0$.



Question 4 | Read parameters from a measured response **Application**

A response is given by

$$x(t) = 0.05e^{-6t}\sin(5t + 1.3333)\ \text{m}.$$

Find the damping ratio, undamped natural frequency, initial displacement, and initial velocity. Is the statement "released from rest" consistent with this response? Can the damping coefficient c be found without knowing the mass?

Solution 4

1. Read the decay and oscillation rates

The exponential rate is $a = \zeta\omega_n = 6\ \text{s}^{-1}$ and the sinusoidal frequency is $b = \omega_d = 5$ rad/s. Therefore

$$\omega_n^2 = a^2 + b^2 = 61, \qquad \boxed{\omega_n = \sqrt{61} = 7.81025\ \text{rad/s}, \quad \zeta = 6/\sqrt{61} = 0.768221.}$$

2. Evaluate the displacement and its derivative

$$x_0 = 0.05\sin(1.3333) = 0.0485965\ \text{m},$$

$$\dot{x} = 0.05e^{-6t}[5\cos(5t + 1.3333) - 6\sin(5t + 1.3333)],$$

$$\boxed{v_0 = -0.232762\ \text{m/s}.}$$

This is not a release from rest. The written response determines the initial conditions; a contradictory verbal description cannot override its derivative.

3. State what is identifiable

$$\frac{c}{m} = 2a = 12\ \text{s}^{-1}, \qquad \frac{k}{m} = 61\ \text{s}^{-2}.$$

Only these ratios are known. If m is supplied, $c = 12m$ and $k = 61m$ in consistent SI units. No unique dimensional c follows from the response alone.

Question 5 | Solve all three response types **Application**

Find $x(t)$ for $x(0) = 0$ and $\dot{x}(0) = 1$ m/s:

$$(a) \quad 2\ddot{x} + 8\dot{x} + 16x = 0,$$
$$(b) \quad 3\ddot{x} + 12\dot{x} + 9x = 0,$$
$$(c) \quad 2\ddot{x} + 8\dot{x} + 8x = 0.$$

Coefficients are in consistent SI units.

Solution 5

1. Underdamped case (a)
Divide by 2: $\ddot{x} + 4\dot{x} + 8x = 0$. Roots are $-2 \pm 2i$, so $x = e^{-2t}(C\cos 2t + D\sin 2t)$. The conditions give $C = 0$ and $2D = 1$:

$$\boxed{x_a(t) = \tfrac{1}{2}e^{-2t}\sin 2t \text{ m.}}$$

2. Overdamped case (b)
Divide by 3: $s^2 + 4s + 3 = (s+1)(s+3) = 0$. Thus $x = Ce^{-t} + De^{-3t}$. The conditions give $C + D = 0$ and $-C - 3D = 1$, hence $C = 1/2, D = -1/2$:

$$\boxed{x_b(t) = \tfrac{1}{2}(e^{-t} - e^{-3t}) \text{ m.}}$$

3. Critically damped case (c)
Divide by 2: $(s+2)^2 = 0$. Use $x = (C + Dt)e^{-2t}$, not two identical exponentials. Then $C = 0$ and $D - 2C = 1$:

$$\boxed{x_c(t) = te^{-2t} \text{ m.}}$$

4. Verify
All three expressions are zero at $t = 0$ and have initial derivative 1. Each tends to zero as $t \to \infty$, but only (a) repeatedly crosses equilibrium.

Question 6 | Overdamping and equilibrium crossing — Challenge

For an overdamped system, prove that a positive displacement-only release remains on the positive side of equilibrium, and that a positive velocity-only release also remains positive for $t > 0$. Explain why this does not establish the same result for every possible pair of initial conditions.

Solution 6

1. Name the two decay rates

Let the roots be $-\alpha, -\beta$, with $0 < \alpha < \beta$. Then $x = Ce^{-\alpha t} + De^{-\beta t}$.

2. Displacement-only release

For $x_0 > 0, v_0 = 0$, solving $C + D = x_0$ and $-\alpha C - \beta D = 0$ gives

$$x = \frac{x_0}{\beta - \alpha}(\beta e^{-\alpha t} - \alpha e^{-\beta t}).$$

Since $e^{-\alpha t} \geq e^{-\beta t}$ for $t \geq 0$, the numerator is positive. Also

$$\dot{x} = \frac{\alpha\beta x_0}{\beta - \alpha}(e^{-\beta t} - e^{-\alpha t}) < 0 \quad (t > 0).$$

It decreases toward zero without crossing.

3. Velocity-only release

For $x_0 = 0, v_0 > 0$,

$$x = \frac{v_0}{\beta - \alpha}(e^{-\alpha t} - e^{-\beta t}) > 0 \quad (t > 0).$$

It initially moves outward, reaches one maximum, and returns asymptotically.

4. Allow general initial data

Now $C = (v_0 + \beta x_0)/(\beta - \alpha)$. If $x_0 > 0$ but $v_0 < -\beta x_0$, then $C < 0$, so the slow term is eventually negative although $x(0) > 0$. A crossing must occur. "Nonoscillatory" does not mean "never crosses for arbitrary initial conditions."

Question 7 | Peak of a critically damped response — Foundation

Derive the time and value of an interior displacement extremum for a critically damped system with initial data x_0, v_0. Then specialize to $x_0 = 0, v_0 > 0$ and identify the positive maximum.

Solution 7

1. Apply the initial conditions

Critical damping gives $x = (C + Dt)e^{-\omega_n t}$. Hence $C = x_0$ and $D = v_0 + \omega_n x_0$, so

$$x = [x_0 + (v_0 + \omega_n x_0)t]e^{-\omega_n t}.$$

2. Locate an interior stationary point

$$\dot{x} = e^{-\omega_n t}[v_0 - \omega_n(v_0 + \omega_n x_0)t].$$

If $v_0 + \omega_n x_0 \neq 0$, a candidate is

$$t_* = \frac{v_0}{\omega_n(v_0 + \omega_n x_0)}.$$

It is relevant only when $t_* > 0$. Substituting back gives

$$x_* = \left(x_0 + \frac{v_0}{\omega_n}\right)\exp\left[-\frac{v_0}{v_0 + \omega_n x_0}\right].$$

The initial point and the limiting value zero must also be considered when asking for a global maximum. Negative initial data can give a minimum instead.

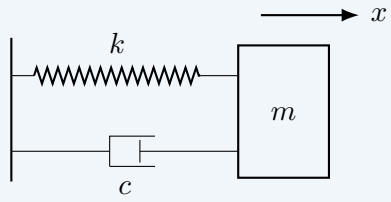
3. Velocity-only release

For $x_0 = 0, v_0 > 0$, $x = v_0 t e^{-\omega_n t}$ and $\dot{x} = v_0 e^{-\omega_n t}(1 - \omega_n t)$. Its sign changes from positive to negative at

$$\boxed{t_{\max} = 1/\omega_n, \qquad x_{\max} = v_0/(e\omega_n).}$$

Question 8 | Critical recoil response — Application

A critically damped recoil mechanism has mass 500 kg and stiffness 10000 N/m. Starting at $x = 0$, its maximum recoil is 0.4 m. Find (a) c_c, (b) its initial recoil velocity, and (c) the time after firing when it returns to $x = 0.1$ m. Take recoil as positive.



Solution 8

1. Determine the frequency and damping

$$\omega_n = \sqrt{10000/500} = 4.472136 \text{ rad/s}, \qquad \boxed{c_c = 2(500)(4.472136) = 4472.14 \text{ N s/m}.}$$

2. Use the maximum recoil

With $x_0 = 0$, $x = v_0 t e^{-\omega_n t}$. Its maximum occurs at $t_p = 1/\omega_n = 0.223607$ s. Therefore

$$0.4 = \frac{v_0}{e\omega_n} \quad \Rightarrow \quad \boxed{v_0 = 0.4e(4.472136) = 4.86261 \text{ m/s}.}$$

The full response is $x(t) = 4.86261te^{-4.472136t}$ m.

3. Select the return crossing

Solve $4.86261te^{-4.472136t} = 0.1$. With $u = \omega_n t$,

$$ue^{-u} = \frac{0.1}{0.4e} = 0.0919699.$$

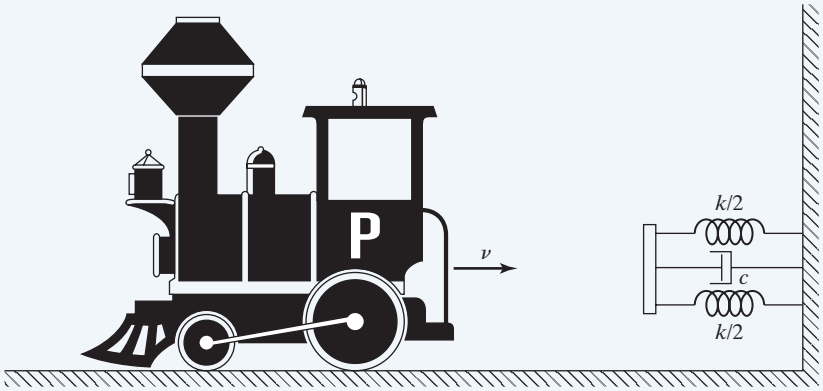
There are two positive roots: one before the peak and one after it. For the return, use $u > 1$. Numerical solution gives

$$\boxed{t_{\text{return}} = 0.825698 \text{ s} \quad \text{after firing.}}$$

This is 0.602091 s after maximum recoil. The mechanism approaches zero asymptotically; it does not reach exact equilibrium in a finite time in this ideal model.

Question 9 | A railroad car and a buffer — Application

A 2000-kg railroad car meets a spring-damper buffer at 10 m/s. The combined stiffness is 80 N/mm and damping is 20 N s/mm. Find the maximum compression and the time to reach it. Set $x(0) = 0$ at first contact.



The two springs together have stiffness k.

Solution 9

1. Convert units and find the response parameters

$$k = 80000 \text{ N/m}, \quad c = 20000 \text{ N s/m}, \quad \omega_n = \sqrt{40}, \quad a = \frac{c}{2m} = 5, \quad b = \sqrt{40 - 25} = \sqrt{15}.$$

Since $a < \omega_n$, motion during compression is underdamped.

2. Impose the contact initial conditions

$$x(t) = \frac{10}{b} e^{-5t} \sin bt.$$

At the first compression maximum, $\dot{x} = 0$, so $b \cos bt - 5 \sin bt = 0$ and

$$\boxed{t_p = \frac{\tan^{-1}(b/5)}{b} = 0.170168 \text{ s}.}$$

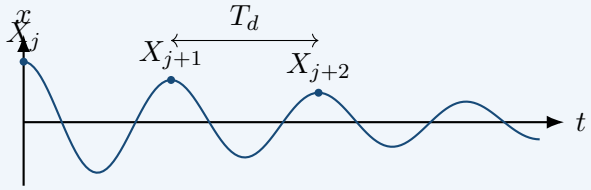
3. Evaluate the compression

$$\boxed{x_{\max} = \frac{10}{\sqrt{15}} e^{-5(0.170168)} \sin(\sqrt{15}\, 0.170168) = 0.675235 \text{ m}.}$$

The free-response model is used while the car and buffer remain in contact. It need not describe the later motion after loss of contact.

Question 10 | Derive logarithmic decrement — Foundation

For an underdamped viscous response, derive the relation between damping ratio and the ratio of two same-sign peaks separated by n full cycles. Explain why adjacent positive and negative peaks cannot be substituted as though they were one full cycle apart.



Schematic: compare peaks on the same side of equilibrium.

Solution 10

1. Compare identical phases

Write $x = Ae^{-at}\cos(bt-\phi)$, $a = \zeta\omega_n$, $b = \omega_d$. Successive same-sign extrema are separated by $T_d = 2\pi/b$. Their trigonometric factors are identical, so

$$\frac{X_j}{X_{j+n}} = e^{anT_d}.$$

Here X_j denotes a positive magnitude of peaks on the same side of equilibrium.

2. Define the per-cycle decrement

$$\delta = \frac{1}{n}\ln\frac{X_j}{X_{j+n}} = aT_d = \frac{2\pi\zeta}{\sqrt{1-\zeta^2}}.$$

Square and rearrange: $\delta^2(1-\zeta^2) = 4\pi^2\zeta^2$. Thus

$$\boxed{\zeta = \frac{\delta}{\sqrt{4\pi^2+\delta^2}}.}$$

3. Distinguish half cycles

Adjacent opposite-sign extrema are separated by $T_d/2$. Their magnitude ratio is $e^{\delta/2}$, not e^{δ}. Taking the logarithm of their signed ratio is also invalid because that ratio is negative.

4. Check measurement practice

Measure displacements relative to equilibrium, count the intervals between peaks, and use well-resolved peaks. Very small late peaks can be dominated by measurement noise. The relation assumes linear viscous damping throughout the record.

Question 11 | Fifty-cycle decay measurement **Application**

A viscously damped body makes five complete oscillations per second. Its amplitude falls to 10% in 50 cycles. Find the logarithmic decrement and damping ratio. By what proportion does the period decrease when damping is removed?

Solution 11

1. Use all 50 cycles

$$\delta = \frac{1}{50} \ln \frac{X_0}{0.1X_0} = \frac{\ln 10}{50} = 0.0460517.$$

2. Recover the damping ratio and frequencies

$$\boxed{\zeta = \frac{0.0460517}{\sqrt{4\pi^2 + 0.0460517^2}} = 0.00732916.}$$

The measured frequency is damped: $f_d = 5$ Hz, $T_d = 0.2$ s, and $\omega_d = 10\pi$ rad/s. Then $\omega_n = \omega_d/\sqrt{1-\zeta^2}$.

3. Compare the periods with the correct denominator

$$T_n = \frac{2\pi}{\omega_n} = T_d\sqrt{1-\zeta^2}.$$

Relative to the original damped period,

$$\boxed{\frac{T_d - T_n}{T_d} = 1 - \sqrt{1-\zeta^2} = 2.68586 \times 10^{-5}.}$$

This is a decrease of 0.00268586%, giving $T_n \approx 0.19999463$ s. A large cumulative amplitude reduction does not require a large frequency change.

Question 12 | Double or halve the damping **Application**

The ratio of successive same-sign amplitudes is 18 : 1 for a viscously damped system. With mass and stiffness unchanged, find the new ratio if the damping coefficient is (a) doubled and (b) halved. Check whether both revised systems still oscillate.

Solution 12

1. Find the original damping ratio

$$\delta = \ln 18 = 2.890372, \qquad \zeta = \frac{\ln 18}{\sqrt{4\pi^2 + (\ln 18)^2}} = 0.417918.$$

2. Change c, keeping c_c fixed

Because m, k are unchanged, $c_c = 2\sqrt{km}$ is unchanged. Thus

$$\zeta_a = 2\zeta = 0.835837 < 1, \qquad \zeta_b = \zeta/2 = 0.208959 < 1.$$

Both systems remain underdamped; the same-sign peak ratio exists for both.

3. Use the exact nonlinear relation

$$R = \exp\left(\frac{2\pi\zeta}{\sqrt{1-\zeta^2}}\right).$$

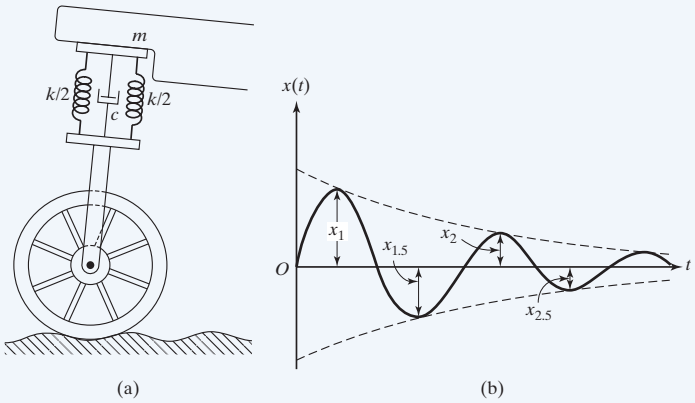
Substitution gives

$$\boxed{R_a \approx 1.42762 \times 10^4, \qquad R_b \approx 3.82886.}$$

Doubling c does not merely double the decrement, because T_d also changes. The small-damping approximation is unsuitable for case (a).

Question 13 | Design a motorcycle shock absorber — Challenge

A motorcycle is modeled by $m = 200$ kg and one equivalent spring and damper. The damped period must be 2 s, and the magnitude of a peak must fall to one-fourth at the next opposite-sign peak. Find k, c. For $x_0 = 0$, find the initial speed that produces a first maximum displacement of 250 mm.



The ratio refers to peak magnitudes separated by half a cycle.

Solution 13

1. Determine decay and oscillation rates

Half a damped period is 1 s. With $a = \zeta\omega_n$, $e^{-a(1)} = 1/4$, so $a = \ln 4 = 1.386294\ \text{s}^{-1}$. Also $b = 2\pi/T_d = \pi$ rad/s.

$$\omega_n = \sqrt{a^2 + b^2} = 3.433863, \qquad \zeta = a/\omega_n = 0.403713.$$

2. Find the physical constants

$$\boxed{k = m\omega_n^2 = 2358.28\ \text{N/m}, \quad c = 2ma = 554.518\ \text{N s/m}.}$$

3. Use the actual peak, not envelope contact

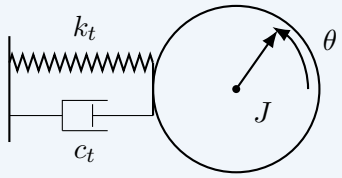
For $x_0 = 0$, $x = (v_0/b)e^{-at}\sin bt$. Differentiate to get $\tan bt_p = b/a$, hence $t_p = 0.367720$ s. At that time $\sin bt_p = b/\omega_n$, giving $x_p = (v_0/\omega_n)e^{-at_p}$.

$$\boxed{|v_0| = 0.250\omega_n e^{at_p} = 1.42926\ \text{m/s}.}$$

For positive v_0, this is the first positive maximum. Later peaks are smaller, so this is also the minimum speed whose response reaches a magnitude of 250 mm.

Question 14 | Identify a dial indicator — Challenge

A torque of 2×10^{-3} N m turns a dial rotor through 50°, corresponding to 80 scale divisions. Released from rest, the pointer reaches -20 divisions after 1 s and $+5$ after another 1 s. Find its torsional stiffness k_t, inertia J, viscous damping c_t, and undamped period.



Solution 14

1. Use static torque and angular displacement

$$k_t = \frac{0.002}{50\pi/180} = 0.002291831 \text{ N m/rad}.$$

Scale divisions are proportional to angle, so their ratios can be used directly.

2. Use the decay and timing

Opposite-sign extrema are 1 s apart, so $T_d = 2$ s and $b = \pi$ rad/s. Their magnitude ratio is $80/20 = 20/5 = 4$, giving $a = \ln 4$ per second. Equivalently, the full-cycle decrement is $\ln(80/5) = \ln 16$.

$$\omega_n = \sqrt{a^2 + b^2} = 3.433863 \text{ rad/s}.$$

3. Recover inertia, damping, and period

$$\boxed{J = \frac{k_t}{\omega_n^2} = 1.94364 \times 10^{-4} \text{ kg m}^2},$$

$$\boxed{c_t = 2Ja = 5.38892 \times 10^{-4} \text{ N m s/rad}},$$

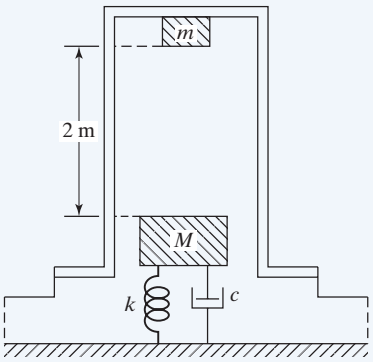
$$\boxed{T_n = 2\pi/\omega_n = 1.82977 \text{ s}.}$$

4. Check the measured sequence

The release-from-rest response is $\theta = \theta_0 e^{-at}[\cos bt + (a/b)\sin bt]$. At $t = 1$, $\theta/\theta_0 = -1/4$; at $t = 2$, it is $1/16$. Thus 80 divisions become -20 and then $+5$, as measured.

Question 15 | Damped response after an impact — Challenge

An anvil weighs 5000 N and rests on a foundation with $k = 5 \times 10^6$ N/m and $c = 10000$ N s/m. A hammer weighing 1000 N falls 2 m onto the initially stationary anvil. The coefficient of restitution is 0.4. Find the anvil response after the first impact, while the bodies remain separated. Measure downward displacement from the anvil's static equilibrium.



(a)

Hammer mass m; anvil mass M.

Solution 15

1. Find the incident velocity and masses

$$u = \sqrt{2gh} = 6.264184 \text{ m/s}, \quad m = 1000/9.81, \quad M = 5000/9.81 \text{ kg}.$$

The short impact is modeled by momentum and restitution; spring and damper impulses are neglected during the collision.

2. Solve the collision equations

$$mu = mv_h + Mv_a, \qquad v_a - v_h = 0.4u.$$

Thus $v_a = (1 + 0.4)mu/(m + M) = 1.461643$ m/s and $v_h = v_a - 0.4u = -1.044031$ m/s. The hammer initially rebounds upward. The anvil has $x_0 = 0$, $v_0 = v_a$.

3. Compute the anvil parameters

$$a = \frac{10000}{2(5000/9.81)} = 9.81 \text{ s}^{-1}, \quad \omega_n = \sqrt{9810} = 99.045444 \text{ rad/s},$$

$$b = \sqrt{9810 - 9.81^2} = 98.558429 \text{ rad/s}.$$

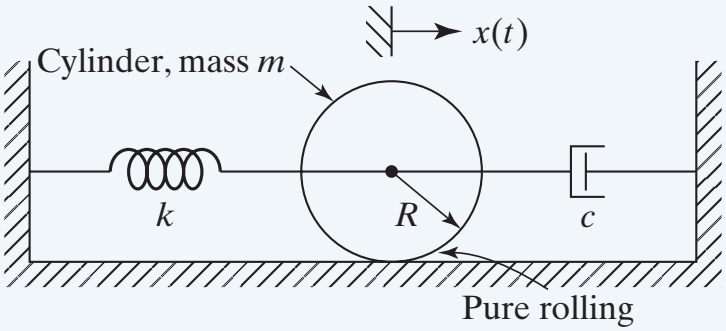
4. Write and check the free response

$$\boxed{x(t) = 0.01483022e^{-9.81t} \sin(98.558429t) \text{ m}.}$$

It gives $x(0) = 0$ and $\dot{x}(0) = 1.461643$ m/s. Do not add the hammer mass to the vibrating mass after separation. This expression ends if a subsequent collision occurs.

Question 16 | A rolling disk with viscous damping — Application

A uniform solid disk of mass m and radius R rolls without slipping. A spring k and viscous damper c act horizontally through its center. Derive the equation in center displacement x and find ω_n and ζ. Explain the role of the contact friction.



The surface provides ideal no-slip rolling.

Solution 16

1. Include translation and rotation

No slip gives $\dot{\theta} = \dot{x}/R$ in magnitude. With $J = mR^2/2$,

$$T = \tfrac{1}{2}m\dot{x}^2 + \tfrac{1}{2}J(\dot{x}/R)^2 = \tfrac{1}{2}\left(m + J/R^2\right)\dot{x}^2 = \tfrac{1}{2}(3m/2)\dot{x}^2.$$

Thus the effective mass is $m_e = 3m/2$. Potential energy is $U = kx^2/2$.

2. Use generalized force

The damping force is $-c\dot{x}$. Ideal rolling friction does no work because the instantaneous contact point has zero velocity. The equation is

$$\boxed{\tfrac{3}{2}m\ddot{x} + c\dot{x} + kx = 0.}$$

3. Read the vibration parameters

$$\boxed{\omega_n = \sqrt{\frac{2k}{3m}}, \quad c_c = 2\sqrt{(3m/2)k}, \quad \zeta = \frac{c}{2\sqrt{(3m/2)k}}.}$$

If $\zeta < 1$, $\omega_d = \omega_n\sqrt{1-\zeta^2}$. Contact friction supplies the torque needed for rolling; it is not automatically a dissipative Coulomb sliding force. The result assumes sufficient static friction to maintain no slip.

Question 17 | Energy during viscous free decay — Foundation

For $m\ddot{x}+c\dot{x}+kx=0$, derive the rate of change of mechanical energy. Find the exact energy ratio at same-sign displacement peaks one cycle apart in terms of logarithmic decrement δ. Is the instantaneous total energy always a constant times $e^{-2\zeta\omega_n t}$?

Solution 17

1. Differentiate the stored energy

$$E=\tfrac{1}{2}m\dot{x}^2+\tfrac{1}{2}kx^2, \qquad \dot{E}=\dot{x}(m\ddot{x}+kx)=-c\dot{x}^2\le 0.$$

The loss rate is zero at a displacement extremum and largest in magnitude where speed is largest. Damping does not cause the energy to increase.

2. Compare displacement peaks

At a turning point $\dot{x}=0$, so $E_j=kX_j^2/2$. Since $X_{j+1}/X_j=e^{-\delta}$,

$$\boxed{\frac{E_{j+1}}{E_j}=e^{-2\delta}, \qquad \frac{E_j-E_{j+1}}{E_j}=1-e^{-2\delta}.}$$

For small δ, $1-e^{-2\delta}\approx 2\delta\approx 4\pi\zeta$. This last relation is an approximation, not an exact identity for large damping.

3. Distinguish a peak sequence from continuous energy

Substitution of $x=Ae^{-at}\cos(bt-\phi)$ and its derivative into E leaves an exponential factor multiplied by a phase-dependent factor. Therefore the exact instantaneous $E(t)$ is generally not a pure exponential. Values separated by a full T_d, however, have identical phase factors and satisfy $E(t+T_d)=e^{-2aT_d}E(t)$.

Question 18 | Loss in a prescribed harmonic cycle **Application**

The prescribed motion is $x(t) = 0.2\sin(\omega_d t)$ m. Find the energy dissipated by a viscous damper in one complete cycle for $m = 10$ kg, $k = 1000$ N/m, and (a) $c = 50$ N s/m, (b) $c = 150$ N s/m. In each case use its underdamped ω_d. Explain why this is not an exact constant-amplitude free response.

Solution 18

1. Integrate damping power

For $x = X\sin\omega t$, $\dot{x} = X\omega\cos\omega t$, so

$$\Delta W = \int_0^{2\pi/\omega} c\dot{x}^2\,\mathrm{d}t = cX^2\omega^2\frac{\pi}{\omega} = \boxed{\pi c\omega X^2}.$$

The average of $\cos^2$ over a full cycle is $1/2$.

2. Find each damped frequency

Here $\omega_n = 10$ rad/s, $c_c = 200$ N s/m. For (a), $\zeta = 0.25$ and $\omega_d = 10\sqrt{1-0.25^2} = 9.682458$ rad/s. For (b), $\zeta = 0.75$ and $\omega_d = 6.614378$ rad/s.

3. Substitute the amplitude in meters

$$\boxed{\Delta W_a = \pi(50)(9.682458)(0.2)^2 = 60.8367\ \text{J}},$$

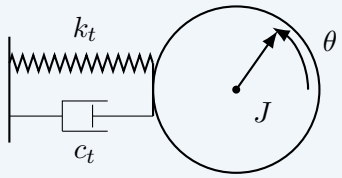
$$\boxed{\Delta W_b = \pi(150)(6.614378)(0.2)^2 = 124.6781\ \text{J}}.$$

4. Interpret the specified motion

A truly free viscously damped response loses amplitude. The question prescribes a constant-amplitude harmonic cycle, which requires energy input to sustain it. This integral is exact for the prescribed cycle; replacing a free-decay cycle with it needs a small-decay approximation.

Question 19 | A torsional oscillator immersed in oil — Application

A disk with inertia $J = 0.2\ \mathrm{kg\,m^2}$ vibrates at 200 cycles/min in a vacuum and 180 cycles/min in oil. Find the torsional damping coefficient. If released from rest in oil at 2°, find its displacement at the end of the first damped cycle. Assume immersion changes damping only.



Solution 19

1. Convert frequencies to rad/s

$$\omega_n = \frac{200(2\pi)}{60} = 20.943951, \qquad b = \omega_d = \frac{180(2\pi)}{60} = 18.849556.$$

2. Recover the decay rate and damping coefficient

$$a = \sqrt{\omega_n^2 - b^2} = 9.129257\ \mathrm{s^{-1}}, \qquad \boxed{c_t = 2Ja = 3.651703\ \mathrm{N\,m\,s/rad}.}$$

The implied damping ratio is $a/\omega_n = \sqrt{1 - 0.9^2} = 0.435890$.

3. Apply a displacement-only release

$$\theta(t) = \theta_0 e^{-at}[\cos bt + (a/b)\sin bt], \quad \theta_0 = 2\pi/180\ \mathrm{rad}.$$

The sine term is needed to make $\dot{\theta}(0) = 0$.

4. Evaluate one full cycle later

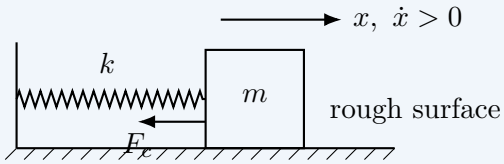
$T_d = 2\pi/b = 1/3$ s, so the sine is zero and cosine is one:

$$\boxed{\theta(T_d) = 2^\circ e^{-9.129257/3} = 0.0953750^\circ.}$$

This remains on the original side. Added inertia or buoyancy-induced stiffness changes would invalidate inference based only on the two measured frequencies.

Question 20 | Derive the Coulomb decrement — Foundation

A mass on a horizontal surface is attached to a spring k. During sliding the friction magnitude is F_c; assume the maximum static friction is also F_c. From a rest position $x(0) = X > F_c/k$, derive the first motion segment and the reduction of peak magnitude per half cycle and per full cycle. State when the decrement rule ceases to apply.



Solution 20

1. Select the friction direction

Initially the mass moves left, so friction acts right. Define $d = F_c/k$ and $\omega_n = \sqrt{k/m}$. Then

$$m\ddot{x} + kx = F_c, \qquad x(t) = d + (X - d)\cos\omega_n t.$$

At $t = \pi/\omega_n$, velocity returns to zero and $x_1 = 2d - X$.

2. Compare alternating extrema

If the next turning point is negative, its magnitude is $X - 2d$. Thus the peak-magnitude loss is $2d = 2F_c/k$ per alternating half cycle. Repeating on the reverse segment gives

$$\boxed{\Delta X_{\text{half}} = 2F_c/k, \qquad \Delta X_{\text{cycle}} = 4F_c/k.}$$

While sliding continues, each full cycle takes $2\pi/\omega_n$; friction shifts the segment equilibrium rather than its frequency.

3. Check sticking before restarting

At any zero-velocity point x_r, the friction needed to remain still is $-kx_r$. Motion can stop whenever

$$\boxed{|kx_r| \le F_c.}$$

If $d < X \le 3d$, the first zero-velocity endpoint lies in this band and the mass sticks there. Do not extrapolate a negative "amplitude" using the decrement formula or force another sign reversal after sticking.

Question 21 | Friction from five measured cycles — Application

A block attached to a spring is released from a displacement of 10 cm on a rough horizontal surface. Five full cycles take 2 s; the same-side peak after those cycles is 1 cm. Find the coefficient of sliding friction. Interpret the 1-cm reading as a measured peak, not necessarily the final resting position.

Solution 21

1. Find the stiffness-to-mass ratio

For continuing Coulomb sliding, $T = 2/5 = 0.4$ s and

$$\omega_n = 2\pi/0.4 = 15.707963 \text{ rad/s}, \qquad k/m = \omega_n^2.$$

2. Use the total amplitude reduction

With normal reaction $N = mg$, each full cycle reduces the amplitude by $4\mu mg/k$. Therefore

$$0.10 - 0.01 = 5\left(\frac{4\mu mg}{k}\right).$$

3. Solve without requiring separate m and k

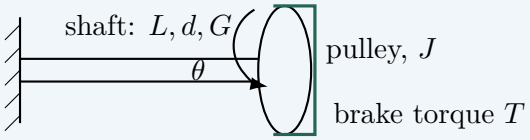
$$\boxed{\mu = \frac{0.09\omega_n^2}{20g} = \frac{0.09(15.707963)^2}{20(9.81)} = 0.11318.}$$

4. Check the wording against static friction

The friction displacement is $d = \mu mg/k = 0.09/20 = 0.0045$ m. With equal static and kinetic coefficients, $0.01 > d$, so the block would not yet remain at rest at that peak. Calling the 1-cm observation the ultimate stopped position would be inconsistent with this interpretation.

Question 22 | A shaft with a friction brake — Application

A solid steel shaft, length 1 m and diameter 50 mm, is fixed at one end and carries a pulley with inertia 25 kg m^2 at the other. A brake provides constant sliding torque 400 N m and the same maximum static torque. Use $G = 8 \times 10^{10}$ Pa. Released from rest at 6°, when and where does the pulley stop?



Solution 22

1. Find torsional stiffness and friction displacement

$$J_p = \pi(0.05)^4/32, \quad k_t = GJ_p/L = 49087.385 \text{ N m/rad},$$
$$d_\theta = 400/k_t = 0.00814873 \text{ rad}, \quad \theta_0 = 6\pi/180 = 0.10471976 \text{ rad}.$$

2. Count completed motion segments

At successive zero-velocity endpoints the signed angle is $\theta_r = (-1)^r(\theta_0 - 2rd_\theta)$ until sticking. The first integer satisfying $|\theta_r| \le d_\theta$ is

$$r = \left\lceil \frac{\theta_0 - d_\theta}{2d_\theta} \right\rceil = \lceil 5.926\ldots \rceil = 6.$$

3. Find final angle and verify the brake can hold it

$$\boxed{\theta_6 = \theta_0 - 12d_\theta = 0.00693496 \text{ rad} = 0.397344^\circ.}$$

It is on the initial side, and $k_t|\theta_6| \approx 340.42$ N m is less than 400 N m.

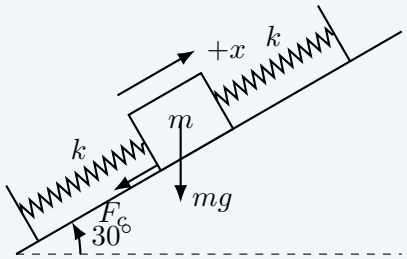
4. Find elapsed time

Each segment takes π/ω_n with $\omega_n = \sqrt{k_t/25}$. Thus

$$\boxed{t_s = 6\pi\sqrt{25/k_t} = 0.425389 \text{ s}, \qquad 3 \text{ full cycles.}}$$

Question 23 | Dry friction on an inclined plane — Challenge

A 20-kg block lies between two springs, each $k = 1000$ N/m, along a 30° incline. Let x be positive upslope from the point where their combined spring force is zero. Take equal static and kinetic coefficients $\mu = 0.1$. Initially $x_0 = 0.1$ m and $v_0 = 5$ m/s upslope. Obtain the response through the first two turning points and give a continuation rule through final sticking.



Friction shown for upslope motion; $x = 0$ is the zero spring-force reference.

Solution 23

1. Keep gravity and friction signs distinct

$$m\ddot{x} + 2kx = -mg\sin 30^\circ - \mu mg\cos 30^\circ \operatorname{sgn}(\dot{x}).$$

Set $\omega_n = \sqrt{2k/m} = 10$ rad/s, $x_e = -mg\sin 30^\circ/(2k) = -0.04905$ m, and $d = \mu mg\cos 30^\circ/(2k) = 0.00849571$ m. Use $y = x - x_e$; its sliding equation is $\ddot{y} + 100y = -100d\operatorname{sgn}(\dot{y})$.

2. First upslope segment

Its shifted equilibrium is $x_p^+ = x_e - d = -0.05754571$ m. Thus

$$x = x_p^+ + 0.15754571\cos 10t + 0.5\sin 10t.$$

The first zero of velocity occurs at

$$t_1 = \frac{\tan^{-1}(0.5/0.15754571)}{10} = 0.126555 \text{ s},$$

$$x_1 = x_p^+ + \sqrt{0.15754571^2 + 0.5^2} = 0.466688 \text{ m}.$$

3. First downslope segment

Now $x_p^- = x_e + d = -0.04055429$ m. For $\tau = t - t_1$,

$$x = x_p^- + (x_1 - x_p^-)\cos 10\tau, \quad 0 \le \tau \le \pi/10.$$

Hence $t_2 = t_1 + \pi/10 = 0.440715$ s and $x_2 = 2x_p^- - x_1 = -0.547796$ m.

4. Continue and stop

At each endpoint set $y_r = x_r - x_e$. If $|y_r| \le d$, stick. Otherwise set $s = \operatorname{sgn}(y_r)$ and use

$$y(t_r + \tau) = sd + (y_r - sd)\cos 10\tau, \quad y_{r+1} = 2sd - y_r.$$

Starting from $y_1 = 0.5157377$ m, 30 further half segments give $y_s = 0.00599513$ m. Therefore $t_s = t_1 + 30\pi/10 = 9.55133$ s and $x_s = -0.0430549$ m. The final check is $|x_s - x_e| < d$.

Question 24 | Find the final resting position **Application**

A 10-kg mass on a horizontal surface is attached to a spring of stiffness 3000 N/m. It is released from rest at $x_0 = 100$ mm. With static and kinetic friction coefficients both equal to 0.12, determine the final position and the time to stop.

Solution 24

1. Calculate the friction force and stopping band

$$F_c = \mu mg = 0.12(10)(9.81) = 11.772 \text{ N}, \qquad d = F_c/k = 0.003924 \text{ m}.$$

Sticking is possible at a zero-velocity point when $|x| \le 3.924$ mm.

2. Track signed endpoints

Starting from $x_0 = 0.100$ m,

$$x_r = (-1)^r(0.100 - 2r \times 0.003924).$$

The number of completed half segments before sticking is

$$r = \left\lceil \frac{0.100 - 0.003924}{2(0.003924)} \right\rceil = \lceil 12.242\ldots \rceil = 13.$$

3. Do not reinterpret a negative residual as another amplitude

$$x_{12} = 0.005824 \text{ m} > d, \qquad \boxed{x_{13} = -(0.100 - 26(0.003924)) = +0.002024 \text{ m}.}$$

The last motion segment ends on the positive side without reaching a negative displacement. The spring needs $3000(0.002024) = 6.072$ N of balancing friction, which is available.

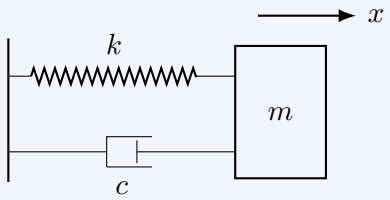
4. Compute elapsed time

$$\boxed{t_s = \frac{13\pi}{\sqrt{3000/10}} \approx 2.35794 \text{ s}.}$$

The half-period count describes intervals between zero velocities; the last interval need not cross equilibrium.

Question 25 | Separate viscous and Coulomb losses — Challenge

A system with possible viscous damping and Coulomb friction loses 1% of a same-sign peak of 20 mm per full cycle, and 2% of a peak of 10 mm per full cycle. Find $F_c/k = \mu N/k$. What do the stated exact measurements imply about its viscous damping component? Assume alternating sliding continues throughout the measured cycles.



The mass also slides on a rough surface: add F_c opposite its velocity.

Solution 25

1. Compare absolute peak losses

Both losses are $0.01(20) = 0.02(10) = 0.2$ mm. A constant absolute decrement suggests Coulomb damping.

2. Derive the combined model's peak map

Let $d = F_c/k$ and $r = e^{-a\pi/b}$, where $a = c/(2m)$ and $b = \sqrt{k/m - a^2}$. Starting from a positive rest peak X, the first negative endpoint is $d - r(X - d)$. Reversing friction for the next half cycle gives the next positive peak

$$X_{\text{next}} = r^2 X - d(1 + r)^2.$$

Thus $X - X_{\text{next}} = (1 - r^2)X + d(1 + r)^2$.

3. Apply both records

In meters,

$$0.0002 = (1 - r^2)(0.020) + d(1 + r)^2,$$
$$0.0002 = (1 - r^2)(0.010) + d(1 + r)^2.$$

Subtracting gives $(1 - r^2)(0.010) = 0$, so $r = 1$ and $a = 0$: the exact data imply zero viscous damping.

$$\boxed{d = 0.0002/4 = 5 \times 10^{-5} \text{ m} = 50\ \mu\text{m}.}$$

If the percentages are rounded experimental values, a small viscous contribution might be unresolved. It is not justified to infer a strictly positive value from these exact numbers.

Question 26 | What does a stopping count tell us? **Challenge**

A 25-N weight hangs from a 1000-N/m spring. It is pulled downward 10 cm from static equilibrium and released. A constant dry-friction force makes it stop after exactly two full cycles (four zero-velocity motion segments). Assuming equal static and sliding friction limits, determine what can be inferred about the friction magnitude. Is it uniquely known?

Solution 26

1. Use the equilibrium coordinate

Gravity is removed by measuring from static equilibrium. Write $X_0 = 0.1$ m and $d = F_c/k$. The signed endpoints before sticking satisfy $x_r = (-1)^r(X_0 - 2rd)$.

2. Enforce both continuation and stopping

To complete four segments rather than stop after three, the third endpoint must lie outside the sticking band. The fourth must lie inside:

$$X_0 > 7d, \qquad X_0 \leq 9d.$$

Equivalently, $3 < (X_0 - d)/(2d) \leq 4$.

3. Give the complete admissible range

$$\frac{X_0}{9} \leq d < \frac{X_0}{7} \quad \Rightarrow \quad \boxed{11.1111 \text{ N} \leq F_c < 14.2857 \text{ N}.}$$

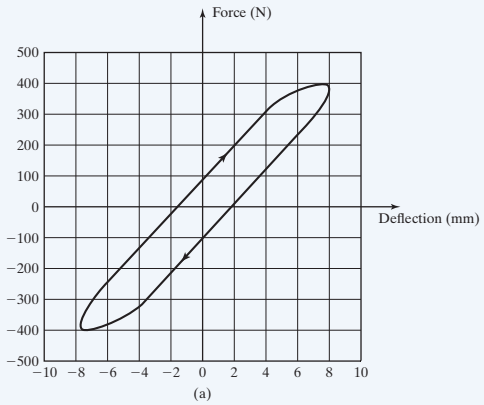
There is no unique force from the count alone. The final position $x_4 = X_0 - 8d$ can be anywhere in the corresponding interval allowed by sticking.

4. Identify the extra datum for a single value

If it were additionally stated that the final position is exactly static equilibrium, then $x_4 = 0$ would give $d = X_0/8$ and $F_c = 12.5$ N. That extra condition is not implied by "comes to rest."

Question 27 | Measure a hysteresis-loop area — Application

A measured force-displacement loop is shown. Each grid square represents 100 N × 2 mm. An area estimate is 12.25 squares. Use peak displacement $X = 8$ mm and stiffness $k = 400/8 = 50$ N/mm to find the energy dissipated per cycle, loss factor $\beta = \Delta W/(\pi k X^2)$, and the small-loss estimate of logarithmic decrement. Discuss that approximation.



Energy loss is the enclosed area, not the area under one branch.

Solution 27

1. Convert the measured area to energy

One square represents 200 N mm = 0.2 J. Thus

$$\boxed{\Delta W = 12.25(0.2) = 2.45 \text{ J}.}$$

The square count is an estimate, so reporting many significant figures would be misleading.

2. Use consistent SI units

$$k = 50000 \text{ N/m}, \quad X = 0.008 \text{ m}, \quad \boxed{\beta = \frac{2.45}{\pi(50000)(0.008)^2} \approx 0.244.}$$

3. Estimate the decay and qualify it

Within the book's nearly harmonic energy balance,

$$\delta = \ln\left(\frac{2+\pi\beta}{2-\pi\beta}\right) \approx 0.807.$$

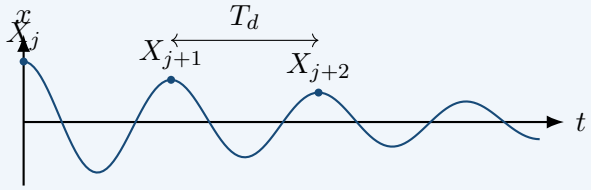
The additional small-loss expansion gives $\delta \approx \pi\beta$, hence

$$\boxed{\delta \approx 0.766, \qquad \zeta_{\text{eq}} \approx \beta/2 \approx 0.122.}$$

Here $E_{\max} = kX^2/2 = 1.6$ J, so the constant-amplitude loop loss divided by $E_{\max}$ is $2.45/1.6 \approx 1.53$. This is not a small-loss loop. The estimate of δ is therefore only a rough extrapolation, not a precise free-decay prediction.

Question 28 | A bridge with structural damping — Application

A bridge is modeled with $m = 5 \times 10^5$ kg and $k = 25 \times 10^6$ N/m. Successive same-sign free-vibration amplitudes have ratio 1.04. Estimate the structural loss factor, equivalent viscous damping and damping ratio, and write an approximate response for arbitrary x_0, v_0.



Schematic: compare peaks on the same side of equilibrium.

Solution 28

1. Start with the book's half-cycle energy balance

Let X and Y be adjacent opposite-sign peak magnitudes. In the nearly harmonic model,

$$\frac{kX^2}{2} - \frac{\pi k\beta}{4}(X^2 + Y^2) = \frac{kY^2}{2}.$$

Thus $(2 - \pi\beta)X^2 = (2 + \pi\beta)Y^2$. The half-cycle ratio is the square root; multiplying two half-cycle ratios gives

$$\boxed{R = \frac{X_j}{X_{j+1}} = \frac{2 + \pi\beta}{2 - \pi\beta}, \qquad \beta = \frac{2(R-1)}{\pi(R+1)}.}$$

This is the unexpanded relation within the book's energy-balance approximation, not an exact constitutive law for arbitrary hysteresis.

2. Use the measured full-cycle ratio

$$\boxed{\beta = \frac{2(1.04 - 1)}{\pi(1.04 + 1)} = 0.01248274.}$$

The logarithmic decrement is $\delta = \ln R = \ln 1.04 = 0.03922071$. For small loss,

$$R \approx 1 + \pi\beta, \qquad \delta \approx \pi\beta, \qquad \zeta_{\text{eq}} \approx \beta/2.$$

3. Find equivalent parameters

$$\omega_n = \sqrt{k/m} = \sqrt{50} = 7.071068 \text{ rad/s}, \quad \zeta_{\text{eq}} \approx 0.00624137,$$

$$\boxed{c_{\text{eq}} \approx \frac{\beta k}{\omega_n} = \beta\sqrt{km} = 44133.15 \text{ N s/m}.}$$

4. Impose the initial conditions

With $a = \zeta_{\text{eq}}\omega_n$ and $b = \omega_n\sqrt{1 - \zeta_{\text{eq}}^2}$,

$$\boxed{x(t) \approx e^{-at}\left[x_0 \cos bt + \frac{v_0 + ax_0}{b} \sin bt\right].}$$

This equivalent viscous response is a small-loss approximation near the natural frequency; it gives $x(0) = x_0$ and $\dot{x}(0) = v_0$.

Question 29 | Composite-panel damping — Application

A composite panel is modeled with $m = 1$ kg, $k = 2$ N/m, and successive same-sign amplitude ratio 1.1. Estimate its hysteretic loss factor β and equivalent viscous coefficient near its natural frequency. Find the harmonic-cycle loss for amplitude $X = 10$ mm.

Solution 29

1. Use the unexpanded peak-ratio relation

For the book's nearly harmonic energy-balance model,

$$R = \frac{2 + \pi\beta}{2 - \pi\beta} \quad \Rightarrow \quad R(2 - \pi\beta) = 2 + \pi\beta.$$

Collect terms: $2(R - 1) = \pi\beta(R + 1)$. With $R = 1.1$,

$$\boxed{\beta = \frac{2(1.1 - 1)}{\pi(1.1 + 1)} = 0.03031523.}$$

The measured decrement is $\delta = \ln 1.1 = 0.09531018$. The simpler estimate $\beta \approx \delta/\pi = 0.03033817$ is close, but is an additional approximation.

2. Match cycle loss near the natural frequency

$$\omega_n = \sqrt{k/m} = \sqrt{2} = 1.414214 \text{ rad/s},$$

$$\boxed{c_{\text{eq}} \approx \frac{\beta k}{\omega_n} = 0.04287221 \text{ N s/m}, \quad \zeta_{\text{eq}} \approx \beta/2 = 0.01515761.}$$

This equivalence matches energy dissipated in nearly harmonic motion at the chosen frequency.

3. Calculate the cycle loss

Convert $X = 10$ mm to 0.010 m:

$$\boxed{\Delta W = \pi\beta k X^2 = \pi(0.03031523)(2)(0.010)^2 = 1.904762 \times 10^{-5} \text{ J}.}$$

4. Distinguish a loop from a decaying record

The peak stored energy is $E = kX^2/2 = 10^{-4}$ J. The constant-amplitude loop estimate gives $\Delta W/E \approx 0.190476$. The exact loss fraction between measured free-decay peaks is $1 - 1/1.1^2 \approx 0.173554$. They differ because the amplitude changes during a free-decay cycle.

Question 30 | Compare damping mechanisms — Foundation

For a prescribed motion of amplitude X and angular frequency ω, compare the cycle losses of (a) a viscous damper c, (b) sliding Coulomb friction F_c, and (c) structural damping with loss factor β. Derive an equivalent viscous coefficient for (b) and (c), and explain why one measured decay curve may not uniquely identify the mechanism.

Solution 30

1. Viscous damping

For $x = X\sin\omega t$,

$$\Delta W_v = \int_0^{2\pi/\omega} c\dot{x}^2\,\mathrm{d}t = \pi c\omega X^2.$$

At fixed amplitude, the loss increases with frequency.

2. Coulomb friction

Over a complete sliding cycle the total distance traveled is $4X$. Therefore

$$\Delta W_C = 4F_cX, \qquad \boxed{c_{\mathrm{eq,C}} = \frac{4F_c}{\pi\omega X}.}$$

Its equivalent coefficient depends on amplitude as well as frequency. This expression presumes full sliding throughout the specified cycle.

3. Hysteretic damping

The loss-factor definition gives

$$\Delta W_h = \pi\beta kX^2, \qquad \boxed{c_{\mathrm{eq,h}} = \frac{\beta k}{\omega}.}$$

If β is approximately frequency independent over the range considered, the loop loss is approximately frequency independent at fixed X.

4. Use more than one signature

Viscous free response has an exponential envelope. Ideal Coulomb sliding gives equal peak decrements and finite sticking. Small-loss hysteretic motion can also show an approximately geometric peak sequence, so an exponential-looking decay alone does not prove viscous damping. Compare force-displacement loops and the dependence on amplitude and frequency. Equivalent coefficients match loss for a specified motion; they do not make the mechanisms identical.